One Capacity, Two Scales: Jamming and the Gravitational Horizon as a Single Event-Density Primitive

Allen Proxmire

June 2026

Series: Event Density (ED) Generative Papers — substrate-evaluation arc (cross-scale capacity note) **Status:** Publication draft. A structural identification across two arcs of the corpus — the soft-matter Universal Mobility Law (Paper_085) and the gravitational-horizon keystone (Phase-3 GR / ED-10). Standalone; cold-reader accessible. **Repository target:** `physics-papers/substrate-evaluation/` (ED-Generative)

> **ON HOLD — UNDER REVISION (2026-06-27).** Being reworked into an EDSC cross-scale *overlay* paper: adding a second worked example — the Local-Group mass sheet Golden-Casimir cavity competing-gradient saddle (ED-SC-01) — alongside the capacity ceiling, and reframed under the two-layer coarse-graining view. Held; **not to be cited as final.** See `event-density/foundations/TwoLayer_CoarseGraining_Hierarchy.md`.

Preamble: What This Paper Does NOT Claim

1. **It does not derive a number bridging the gel and the black hole.** The two phenomena live ~40 orders of magnitude apart. The claim is that they are governed by the *same primitive in the same form*, not that one is computed from the other.
2. **It does not re-validate either arc.** The soft-matter degenerate mobility is empirically validated independently (eleven systems, Paper_085); the gravitational horizon is form-derived in the GR keystone (khronometric metric, area-law and Hawking-temperature scalings). This note identifies the shared primitive between two established results; it adds to neither.
3. **The 13 substrate primitives are not derived** (Paper_087). This is a structural identification within ED, not a derivation of it.
4. **"Capacity" means the P04 participation-bandwidth bound.** The claim is about that one primitive; no new primitive is introduced.

Abstract

Two results in the Event-Density corpus, developed in unrelated arcs, turn on the same primitive: the finite participation-bandwidth capacity, P04. In **soft matter**, the Universal Mobility Law M( ) = M (_max −)^ — validated across eleven systems — says transport halts as density approaches the capacity _max: a gel jams when its participation is full. In **gravity**, the keystone result says the metric degenerates as the available bandwidth collapses, g ~ b ¹ → ∞ as b → 0: a horizon forms when participation can no longer propagate. These are the same event — capacity exhaustion stopping propagation — seen from reciprocal sides: as a mobility that *vanishes* (soft matter) and as a metric that *diverges* (gravity), since mobility ~ (capacity − occupancy) and the metric ~ (bandwidth) ¹. The identification predicts a shared *form* —

propagation response degenerating as a power of (capacity − occupancy) — across the two scales, and it is falsifiable: structurally different capacity laws for jamming and for horizon formation would refute it. The companion channel-ledger paper makes the link concrete: the capacity that the bare worldline substrate lacks (so that it gives only *linear* diffusion, not the degenerate Universal Mobility Law) is exactly the $b \to 0$ bandwidth collapse that the gravity arc uses to form horizons. The capacity that makes a crowded gel stop flowing is the capacity that makes a black hole.

---

1. Introduction

The Event-Density corpus reaches two phenomena by long, separate routes. One arc, in soft matter, arrives at a mobility law for crowded media. Another, in gravity, arrives at the horizon. This note observes that the two arrive at the *same primitive* — the finite participation capacity P04 — and that the soft-matter law and the gravitational horizon are two faces of capacity exhaustion. The observation is structural and narrow: it does not unify the scales numerically, and it does not strengthen either result. What it does is locate a single mechanism beneath two phenomena that look unrelated, and state the cross-scale prediction that follows.

2. The capacity primitive

P04 gives each region of the substrate a finite participation bandwidth — a ceiling on how much becoming it can carry and pass on. Two consequences of approaching that ceiling are central here:

- **Transport saturates.** As a region's occupancy approaches its capacity, its ability to accept and forward further participation falls; at the ceiling, transport halts.
- **The metric degenerates.** Where the available bandwidth collapses to zero, the kinematic metric — which scales inversely with bandwidth, $g \sim b^{1}$ — diverges, and propagation becomes one-way.

These are the same fact about a finite channel, expressed as a vanishing throughput and as a diverging impedance.

3. The soft-matter face: degenerate mobility

The Universal Mobility Law M( ) = M (_max −)^ describes transport in crowded soft matter: the mobility is a decreasing function of density that **vanishes at the capacity _max**. Approaching maximum packing, motion stops — the medium jams. The law is validated across eleven chemically distinct systems (Paper_085), with the degeneracy at _max its defining, empirically anchored feature. In substrate terms, _max is the local participation capacity, and the vanishing mobility is transport saturating as that capacity fills.

4. The gravitational face: bandwidth collapse

The gravity keystone derives a khronometric metric whose strength scales inversely with the available participation bandwidth, with a horizon where that bandwidth collapses: as $b \to 0$, the metric degenerates and the region self-isolates (one-way participation). On the resulting horizon the keystone delivers the expected thermodynamics — an area law and a Hawking-type temperature scaling. The horizon is, structurally, the place where participation capacity has been fully committed and can carry no more.

5. The identity

Set the two faces side by side. In soft matter the observable is the **mobility**, M ~ (_max −)^, which **vanishes** as occupancy approaches capacity _max. In gravity the observable is the **metric**, $g \sim b^{1}$, which **diverges** as bandwidth b approaches zero. Occupancy approaching capacity (→ _max) is bandwidth approaching zero ($b \to 0$): a full channel. Vanishing mobility and diverging metric are reciprocal descriptions of the same saturation — throughput $\to 0$ is impedance $\to \infty$.

So a single primitive, P04 capacity, casts two phenomena: the gel that stops flowing at maximum packing, and the horizon that forms where bandwidth runs out. The channel-ledger companion paper sharpens the link from the substrate side: the certified worldline rule, lacking a capacity bound, coarse-grains only to *linear* diffusion, never the degenerate Universal Mobility Law — the degenerate mobility appears precisely when the capacity (the b $\rightarrow$ 0 bound, the same one the gravity arc uses) is supplied. The missing ingredient on the soft-matter side and the horizon-forming mechanism on the gravity side are the same primitive.

6. What it predicts

The identification is not decorative; it carries a cross-scale prediction and a way to fail. It predicts that the **form** of capacity saturation is shared — propagation response degenerating as a power of (capacity − occupancy), transport $\rightarrow$ 0 / impedance $\rightarrow \infty$ — at both scales, because both are the one P04 ceiling. It is **falsifiable**: if jamming and horizon formation were governed by structurally different capacity laws — one a power-law degeneracy in (capacity − occupancy) and the other not, or with no shared (capacity − occupancy) dependence — the single-primitive claim would be refuted. The scales differ by tens of orders of magnitude; what is asserted to be invariant is the primitive and its functional form, not the values.

7. Conclusion

Two of the corpus's results — the soft-matter mobility law and the gravitational horizon — are two faces of one Event-Density primitive: the finite participation capacity P04. A crowded medium jams when occupancy reaches capacity; a horizon forms when bandwidth reaches zero; these are the same channel filling, read as a vanishing mobility and as a diverging metric. The claim is structural and cross-scale, falsifiable through the shared degenerate form, and it ties the program's most empirically anchored law to its gravitational keystone through a single mechanism. The capacity that makes a crowded gel stop flowing is the capacity that makes a black hole.